

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 05



Submitted to:

Sir Hassan Shah

Submitted by:

Zulqarnain Hassan

Roll No: SP21379

Subject: computer network

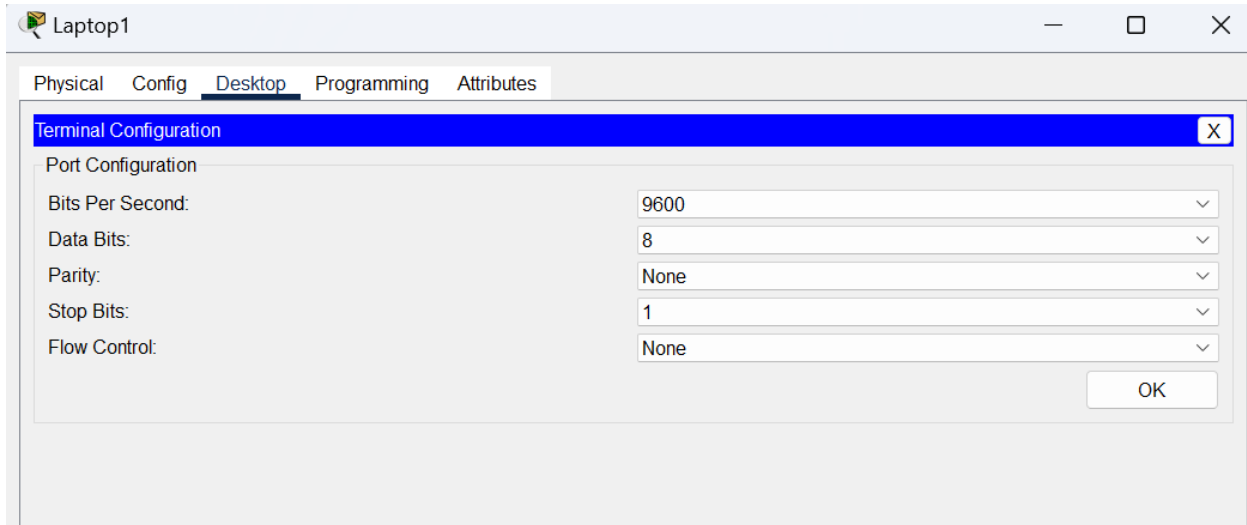
Department: BSSE-5-Afternoon

Date: 19-11-2024

a. (Switch)

Task 1

Click on pc2 > click desktop tab > click on terminal > click ok (you should be able to access a switch or a Router)



Task 2:

MOTD:

```
Switch>en
Switch#
Switch#
Switch#confi
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#banner motd #....admin access only....#
Switch(config)#
```

Task 3:

SECURING CONSOLE AND TELNET PORTS:

```
Zulgarnain_Hassan-SP21379(config)#
Zulgarnain_Hassan-SP21379(config)#line vty 0 15
Zulgarnain_Hassan-SP21379(config-line)#
Zulgarnain_Hassan-SP21379(config-line)#password lab-5
Zulgarnain_Hassan-SP21379(config-line)#
Zulgarnain_Hassan-SP21379(config-line)#exit
Zulgarnain_Hassan-SP21379(config)#line con 0
Zulgarnain_Hassan-SP21379(config-line)#password lab-5line-0
Zulgarnain_Hassan-SP21379(config-line)#logging synchronous
Zulgarnain_Hassan-SP21379(config-line)#login
Zulgarnain_Hassan-SP21379(config-line)#
Zulgarnain_Hassan-SP21379(config-line)#histo
Zulgarnain_Hassan-SP21379(config-line)#history size 20
Zulgarnain_Hassan-SP21379(config-line)#exec-time
Zulgarnain_Hassan-SP21379(config-line)#exec-timeout 5 30
Zulgarnain_Hassan-SP21379(config-line)#
```

Task 4:

Set Privileged EXEC password:

```
Zulqarnain_Hassan-SP21379(config)#enable
Zulqarnain_Hassan-SP21379(config)#enable secret lab-5
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#
```

Task 5:

Enable SSH:

```
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#ip domain-name CN-lab5
Zulqarnain_Hassan-SP21379(config)#crypto key generate rsa
The name for the keys will be: Zulqarnain_Hassan-SP21379.CN-lab5
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 800
% Generating 800 bit RSA keys, keys will be non-exportable...[OK]

Zulqarnain_Hassan-SP21379(config)#
*Mar 1 0:12:27.737: %SSH-5-ENABLED: SSH 1.99 has been enabled
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#
```

Task 6

Setting SSH version to 2:

```
Zulqarnain_Hassan-SP21379(config)#ip ssh version 2
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#line vty 0 5
Zulqarnain_Hassan-SP21379(config-line)#transport input ssh
Zulqarnain_Hassan-SP21379(config-line)#
Zulqarnain_Hassan-SP21379(config-line)#login local
Zulqarnain_Hassan-SP21379(config-line)#login local
Zulqarnain_Hassan-SP21379(config-line)#user
Zulqarnain_Hassan-SP21379(config-line)#username Zulqarnain_Hassan-SP21379 password vty05
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#
```

Task 7

Switch Port Access:

```
Zulqarnain_Hassan-SP21379(config)#
Zulqarnain_Hassan-SP21379(config)#in
Zulqarnain_Hassan-SP21379(config)#interface fa
Zulqarnain_Hassan-SP21379(config)#interface fastEthernet 0/2
Zulqarnain_Hassan-SP21379(config-if)#switchport mode access
Zulqarnain_Hassan-SP21379(config-if)#switchport port-security mac-address sticky
^
% Invalid input detected at '^' marker.

Zulqarnain_Hassan-SP21379(config-if)#switchport port-security mac-address sticky
Zulqarnain_Hassan-SP21379(config-if)#switchport port-security maximum 1
Zulqarnain_Hassan-SP21379(config-if)#
Zulqarnain_Hassan-SP21379(config-if)#
```

Task 8

Switch Port Security:

```
Zulqarnain_Hassan-SP21379(config-if)#
Zulqarnain_Hassan-SP21379(config-if)#
Zulqarnain_Hassan-SP21379(config-if)#switchport port-security violation shutdown
Zulqarnain_Hassan-SP21379(config-if)#
Zulqarnain_Hassan-SP21379(config-if)#
```

b. (Router)

Task 1:

connect to the Router using the console cable.

Physical Config **Desktop** Programming Attributes

Terminal Configuration [X]

Port Configuration

Bits Per Second:	9600
Data Bits:	8
Parity:	None
Stop Bits:	1
Flow Control:	None

OK

Task 2

MOTD

```
Router>en
Router#conf
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#banner motd #....massege of the day....#
Router(config)#
```

Task 3

Specifies a virtual terminal for remote console access

```
Router(config)#banner motd #....massege of the day....#
Router(config)#line vty 0 15
Router(config-line)#password lab-5
Router(config-line)#login
Router(config-line)#
```

Task 4

Sets the interval that the EXEC command interpreter waits until user input is detected. The default is 10 minutes.

```
Router(config-line)#login
Router(config-line)#exec-
Router(config-line)#exec-timeout 10
Router(config-line)#
```

Task 5

```
Router(config)#inter
Router(config)#interface fas
Router(config)#interface fastEthernet 0/0
Router(config-if)#description NUML-SE-LAB-3
Router(config-if)#
```

Task 6

```
....massege of the day....
Router>
Router>en
Router#conf
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable se
Router(config)#enable secret cisco
Router(config)#
Router(config)#ser
Router(config)#service pass
Router(config)#service password-encryption
Router(config)#
```